

Probability

Random Variables, Probability Distributions

Sampling

A **Simple Random Sample** (SRS) of size n from a finite population is a subset selected in such a way that every subset of n elements is equally likely to be the one selected.

Random Variables

A **random variable** is a function defined on a numeric population. We often say that this function value is the value assumed by the random variable. A probability distribution specifies the probabilities associated with all possible values the random variable can assume.

Example

If X represents the number of RED *M&M*s in a vending machine size package, then perhaps the probability distribution of X is something like the following:

x	5	6	7	8	9	10	11
$P(X = x)$	$\frac{2}{28}$	$\frac{2}{28}$	$\frac{5}{28}$	$\frac{7}{28}$	$\frac{5}{28}$	$\frac{4}{28}$	$\frac{3}{28}$

Discrete Random Variables

When the elements of a population consists of a countable discrete set of values the random variable defined on that population is called discrete random variable.

Properties:

- $0 \leq P(X = x)$ for all x
- $\sum_x P(X = x) = 1$

Continuous Random Variables

When the population consists of a set of values in a continuous range the random variable defined on that population is called continuous random variable. The probability distribution of a continuous random variable is a smooth curve, and the area under the curve between specified values of the random value correspond to a probability.

Continuous Random Variables

Properties:

- $0 \leq f(x)$ for all x
- $\int f(x)dx = 1$

Continuous RV (example)

Suppose that our population consists of all numbers in the interval $(0,2)$ occurring with various frequencies as specified by $f(x)$.

- $P(1/2 \leq X \leq 1/2) =$

- $P(0 \leq X \leq 2) =$

- $P(1 \leq X \leq 2) =$

Common Distributions

For some interesting populations that occur naturally, theoretical probability distributions and associated random variables, have been defined.

Discrete

- Binomial
- Poisson
- Negative Binomial

Continuous

- Normal
- Gamma
- Chi squared
- F
- t

Binomial Experiment

A binomial experiment consists of taking n successive SRS, each of size 1, replacing elements between draws, and counting the number of “successes” (S) obtained, m . The alternative is “failures” (F).

- $\pi = m/n$
- $1 - \pi = (N - m)/N$

Let X_i be the random variable that assumes the value of the outcome of the i -th draw for each $i = 1, 2, \dots, n$.

Binomial Distribution

Let X_i assume the value 1 if the outcome of the i -th draw is S or the value of 0 if the outcome is F. For a repeated process, n random variables, $X_1, X_2, \dots, X_n$, are distributed as follows:

x	0	1
$P(X = x)$	$1 - \pi$	π

Binomial Distribution (contd.)

Now define a new random variable Y so that

$$Y = X_1 + X_2 + X_3 + \cdots + X_n$$

- Y represents the number of “successes” for the n random variables
- Y is said to be a binomial random variable.

Binomial Distribution (contd.)

$$Y \sim \text{Binomial}(n, \pi); \quad y = 1, 2, \dots, n; \quad 0 < \pi < 1$$

$$P(Y = y) =$$

Mean and Variance of a Discrete Random Variable

- The **mean** of a discrete random variable X is:

$$E(X) = \mu = \sum_x x \cdot P(X = x)$$

- The **variance** of a discrete random variable X is:

$$V(X) = \sigma^2 = \sum_x (x - E(X))^2 \cdot P(X = x)$$

- The **standard deviation** of a random variable X is

$$\sqrt{V(X)} = \sigma$$

Mean and Variance of a Discrete Random Variable (contd.)

- Using the previous formula, the **mean** of a Binomial random variable Y , is shown to be $E(Y) = \mu = n\pi$
- The **variance** is shown to be $V(Y) = \sigma^2 = n\pi(1 - \pi)$.
- Thus the **standard deviation** of a Binomial random variable is $\sigma = \sqrt{n\pi(1 - \pi)}$

Binomial Distribution: Example

Consider a Binomial population where $Y \sim \text{Bin}(3, 1/3)$.

- Mean:
- Variance:
- Standard Deviation:

Assumptions of the Binomial Model

- . n identical trials are made.
- . Each trial results in either **Success(S)** or **Failure(F)**. (only two possible outcomes).
- . The probability of success (π) remains the same from trial to trial.
- . The trials are independent.
- . Interest is in the number of successes obtained in the n trials. The number is viewed as being the value assumed by a Binomial (n, π) random variable Y .

Binomial Distribution: R

Let $Y \sim \text{Bin}(n, \pi)$

- $P(Y = y)$ is given by `dbinom(x=y, size=n, prob=pi)`
- $P(Y \leq y)$ is given by `pbinom(x=y, size=n, prob=pi)`

Mean and Variance of a Continuous Random Variable

- The **mean** of a continuous random variable X is:

$$E(X) = \mu = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

- The **variance** of a continuous random variable X is:

$$V(X) = \sigma^2 = \int_{-\infty}^{\infty} (x - E(X))^2 \cdot f(x) dx = E(X^2) - E(X)^2$$

- The **standard deviation** of a random variable X is

$$\sqrt{V(X)} = \sigma$$

Normal Random Variable

The Normal probability distribution is an example of a continuous distribution, and is defined in the interval $(-\infty, \infty)$. This is a bell-shaped distribution and is commonly used to model continuous data.

$$f(x) =$$

where

- $E(X) =$

- $V(X) =$

- $\sqrt{V(X)} =$

Properties of the Normal Distribution

- Symmetric about the mean μ
- Probabilities calculated as

$$P(a \leq X \leq b) = \frac{1}{\sqrt{2\pi}\sigma} \int_a^b e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

Standard Normal Distribution

The standard normal distribution is a normal distribution with $\mu = 0$ and $\sigma^2 = 1$, and is represented by the random variable Z . If $X \sim N(\mu, \sigma^2)$, then the following transformation converts X into a standard normal random variable.

$$Z = \frac{X - \mu}{\sigma}$$

Standard Normal Distribution

If $X \sim N(0, \sigma^2)$, then we can use a Z-table to compute areas under the curve. For example,

$$P(a \leq X \leq b) =$$

Standard Normal Distribution: R

We may also use R to compute probabilities for random variables that follow a normal distribution. While any μ and σ can be selected, the default is 0 and 1, respectively.

- `dnorm()`: computes $f(z)$
- `pnorm()`: computes p from $P(Z \leq z) = p$ for a specified z
- `qnorm()`: computes z from $P(Z \leq z) = p$ for a specified p

Standard Normal Distribution: Examples

. $P(Z \geq 0) = P(Z \leq 0) = 0.50$

```
1-pnorm(0); pnorm(0)
```

```
[1] 0.5
```

```
[1] 0.5
```

. $P(0 \leq Z \leq 0.12) = P(Z < 0.12) - P(Z < 0) = 0.5478 - 0.5 = 0.0478$

```
pnorm(0.12) - pnorm(0)
```

```
[1] 0.04775843
```

. $P(-0.12 \leq Z \leq 0.12) = 2 \cdot P(0 \leq Z \leq 0.12) = 0.0956$

```
pnorm(0.12) - pnorm(-0.12)
```

```
[1] 0.09551685
```

Standard Normal Distribution: Examples

- Find z such that $P(0 \leq Z \leq z) = 0.4846$

```
qnorm(p = (0.4846 + pnorm(0)))
```

```
[1] 2.159647
```

- Find z such that $P(z \leq Z \leq 0) = 0.2257$

```
qnorm(p = -(0.2257 - pnorm(0)))
```

```
[1] -0.5998593
```

- Find z such that $P(Z \geq z) = 0.011$

```
qnorm(0.011, lower.tail = F)
```

```
[1] 2.290368
```

Normal Distribution

Similar to the previous problems, if we know μ and σ , we can solve probability problems using the `mean` and `sd` arguments in the `qnorm()` functions.

```
qnorm(0.975, mean = 5, sd = 0.2)
```

```
[1] 5.391993
```

Sampling Distribution of the Sample Mean $\bar{Y}$

Define the Sample Mean as

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

where $Y_1, Y_2, \dots, Y_n$ are taken independently from the population. That is, taking the sum of all of the values and dividing it by the count of values. Any function of $Y_1, Y_2, \dots, Y_n$ is a random variable, including $\bar{Y}$.

Properties of the Sample Mean $\bar{Y}$

If $Y_1, Y_2, \dots, Y_n$ are each independently distributed from a normal distribution with mean μ and variance σ^2 , then

- $E(\bar{Y}) = \mu$
- $V(\bar{Y}) = \sigma^2/n$

Standard Error of the Mean: the standard deviation of $\bar{Y}$

Central Limit Theorem

When sample size n is large, the random variable $\bar{Y}$ is approximately normally distributed with mean μ and variance σ^2/n . That is,

$$\bar{Y} \approx N(\mu, \sigma^2/n)$$

Central Limit Theorem: Simulation

Let's take $n = 20$ and let $Y \sim N(50, 16)$. A single sample of size 20 from this population may look like the following:

```
set.seed(313)
y <- rnorm(n = 20, mean = 50, sd = sqrt(16))
head(y)
```

```
[1] 49.99007 52.82391 53.15559 45.64610 53.38200 51.13121
```

```
mean(y)
```

```
[1] 50.0067
```

```
var(y)
```

```
[1] 10.63239
```

```
sd(y)
```

```
[1] 3.260734
```


Central Limit Theorem: Simulation

Let's say that we take many samples of size 20 from the population. We would expect $\bar{Y}$ to be approximately normal with mean 50 and variance $16/20 = 0.8$.

- `replicate()`: repeat an expression n times (different n from `rnorm`)
- `apply()`: for each column (`MARGIN = 2`), apply the mean function

```
many_samples <- replicate(n = 1000, rnorm(n = 20, mean = 50, sd = 3))
y_bar <- apply(many_samples, MARGIN = 2, mean)
hist(y_bar)
```

Central Limit Theorem: Simulation

Let's say that we take many samples of size 20 from the population. We would expect $\bar{Y}$ to be approximately normal with mean 50 and variance $16/20 = 0.8$.

- `replicate()`: repeat an expression n times (different n from `rnorm`)
- `apply()`: for each column (MARGIN = 2), apply the mean function

```
mean(y_bar)
```

```
[1] 49.97217
```

```
var(y_bar)
```

```
[1] 0.4365369
```

```
sd(y_bar)
```

```
[1] 0.6607094
```

Central Limit Theorem: Example

Suppose that you are estimating the average weight of sheep in a large herd. You obtain a random sample of size $n = 50$, and can assume that $\mu = 40$ and $\sigma^2 = 4$. What is the probability that the sample mean will be within 0.1 pounds of μ ?

$$\bar{Y} \approx$$

Central Limit Theorem: Example

$$P(39.9 \leq \bar{Y} \leq 40.1)$$

Standard Normal Distribution

```
pnorm((40.1-40) / sqrt(4/50)) - pnorm((39.9-40) / sqrt(4/50))
```

```
[1] 0.2763264
```

Normal Distribution

```
pnorm(40.1, mean = 40, sd = sqrt(4/50)) - pnorm(39.9, mean = 40, sd = sqrt(4/50))
```

```
[1] 0.2763264
```

Poisson Distribution

Poisson distribution is used to model the number of occurrences of “rare” events in certain intervals of time and space. A Poisson random variable is a discrete random variable as it represents a count.

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, 3, \dots$$

where λ is the mean parameter and represents the expected number of events occurring in the interval of interest.

Poisson Distribution Examples

- X = # of alpha particles emitted from a polonium bar in an 8 minute period
- Y = # of flaws on a standard size piece of manufactured product (e.g., 100m coaxial cable, 100 sq.meter plastic sheeting)
- Z = # of hits on a web page in a 24h period

Poisson Distribution Properties

Let $X \sim Poi(\lambda)$.

- $E(X) =$
- $V(X) =$
- If $X_i \sim Poi(\lambda)$ for $i = 1, 2, \dots, n$, then $\sum_{i=1}^n X_i \sim Poi(n\lambda)$

Poisson Distribution: R

Let $X \sim Poi(\lambda)$.

- $P(X = x)$: `dpois(x=x, lambda=lambda)`
- $P(X \leq x)$: `ppois(x=x, lambda=lambda)`

Example problems

Let $\mu = 10$ and $n = 30$. When required, approximate σ^2 using the variance formula of the Binomial distribution. Compute each of the following problems for the Binomial, Normal, and Poisson distributions, as well as for $\bar{Y}$.

- $P(X \leq 4)$
- $P(X < 4)$
- $P(12 \leq X \leq 24)$
- x such that $P(X \leq x) = 0.025$
- x such that $P(X > x) = 0.0975$

Extra Mean/Varianc Properties

- $E(X) = \int_x x \cdot f(x) dx$
- $E(aX + B) = aE(X) + b$
- $V(X) = E((X - \mu)^2) = E(X^2) - E(X)^2$
- $V(aX + B) = a^2 V(X)$

Stat 801A

Speaker notes